University Course Timetabling

STATE OF THE ART AND REPRESENTATIVE METHODS FOR CURRICULUM-BASED COURSE TIMETABLING

PROBLEM AND OBJECTIVE

University timetabling assigns lectures to periods and rooms under teacher, room, and curriculum constraints. Timetable quality also depends on room capacity, weekly distribution, curriculum compactness, and room stability. This study focuses on **Curriculum-Based Course Timetabling (CB-CTT)** and the public UD2 formulation from ITC-2007. It will review current methods and compare a small set of representative implementations.

SCOPE OF THE REVIEW

The review will cover CB-CTT research published from 2015 to September 2026, plus the foundational work behind the benchmark. Searches in Scopus, Web of Science, and IEEE Xplore will be extended through citation tracing. Each paper will be classified by formulation, dataset, method, evidence, computational setting, and code availability. The analysis will examine how each family builds feasible timetables, improves their quality, and uses information about the instance. Work on other timetabling variants will be included when the method transfers to CB-CTT.

FAMILY	REVIEW COVERAGE	RECENT DIRECTION
Exact and constraint	MIP, CP, CP-SAT, decomposition, bounds	Model-based construction and decomposition
Metaheuristics	SA, tabu search, VNS/VND, ALNS, hyper-heuristics	LectureKick Simulated Annealing [5]
Hybrid methods	Solver-assisted construction and local improvement	CP-SAT initialization followed by VND [6]
Learning-assisted	Feature-based tuning, algorithm selection, adaptive control	Instance-space and algorithm-selection studies [3]

PLANNED IMPLEMENTATION

We intend to implement CP-SAT, Simulated Annealing, and a CP-SAT plus VND hybrid informed by Nguyen [6]. CP-SAT provides a direct model of the constraints and objective. SA represents stochastic local search, while the hybrid combines solver-based construction with neighbourhood improvement. Evaluation will use the **19 non-redundant ITC-2007 UD2 instances**. Instance `comp11` is omitted because competitive methods consistently reach cost zero. Instance `comp15` duplicates `comp03` under UD2 [1]. The comparison will report feasibility, objective value, and running time. Its final design will be fixed after the review and initial trials.

EXPECTED CONTRIBUTION AND SCOPE

The final article will connect the literature review with results from the three implementations. It will identify their practical differences and place them alongside recent CB-CTT work. Learning-assisted approaches will remain part of the review.

REFERENCES

[1] Ceschia, Di Gaspero, and Schaerf, "Educational timetabling: Problems, benchmarks, and state-of-the-art results," 2023. [2] Abdipoor et al., "Metaheuristic approaches for university timetabling problems," 2023. [3] De la Rosa-Rivera et al., "Algorithm selection for educational timetabling," 2021.

[4] Gu et al., "From integer programming to machine learning: A survey on solving university timetabling problems," 2025. [5] Ceschia et al., "Revisiting local search for curriculum-based course timetabling," 2026. [6] Nguyen, "Adaptive penalty-guided VND with CP-SAT for curriculum-based course timetabling," 2026. [7] Di Gaspero, McCollum, and Schaerf, *The Second International Timetabling Competition (ITC-2007): Curriculum-Based Course Timetabling*, 2007.